

Slicer + Compare Engines Tutorial

Send to OrcaSlicer in two clicks. Run two engines side-by-side. Pick the one that prints faster, lighter, or stronger.

ForgeSlicer ships with a built-in OrcaSlicer engine (Flatpak distribution on Linux/ARM64, native on macOS/Windows). You can also send the same 3MF to your desktop OrcaSlicer / Bambu Studio / PrusaSlicer with one click.

1. Engines, profiles, and the slice pipeline

A **slicer engine** is the program that converts your 3D mesh into G-code — layer-by-layer instructions a printer actually understands. ForgeSlicer routes slicing requests to one of:

- **OrcaSlicer (built-in)** — runs server-side on ForgeSlicer's CLI. Default. Best for one-click slicing.
- **Desktop slicer (sent file)** — downloads a 3MF you open manually in your desktop slicer of choice. Best when you want full control of advanced settings.

Profile inheritance

ForgeSlicer profiles inherit from upstream OrcaSlicer system profiles — same syntax, same parameter names, same results. Your custom overrides live in the right panel (**Printer**, **Filament**, **Process**) and are applied as a delta on top of the chosen base profile.

2. Send to Slicer — the main flow

When your design is ready, click **Send to OrcaSlicer** in the top toolbar. A dialog appears with four sections:

Section	What it controls
Printer	Bed size, nozzle diameter, max print height. Pick a system profile or a saved custom one.
Filament	Material (PLA/PETG/ABS/TPU/PA-CF), print temp, bed temp, flow ratio. Quickly compare materials — PLA vs PETG can change your part's print time by 20%.
Process	Layer height, walls, infill, supports, top/bottom layers. Most quality-vs-speed tradeoffs live here.
Engine	Pick OrcaSlicer (server) or download a 3MF for a desktop slicer. The right panel switches accordingly.

Slicing on the server

- 1 Click **Slice**. Progress streams to the dialog in real time (loading mesh → perimeters → infill → supports → G-code).
- 2 When done, a **Slice Summary** appears: estimated print time, filament length, filament weight, peak nozzle temperature, and bed level deviation.
- 3 Click **Download G-code** to save the file, or **Preview G-code** to scrub through the layers in a viewer overlay.

Slow first slice? The Flatpak runtime is downloaded lazily the first time you click Slice on an ARM64 host. Subsequent slices use the cached runtime and start in <1 s.

3. Compare Engines — A/B your slices

Compare Engines runs the same design through two slicer configurations and lays the results side-by-side. Use it to answer questions like:

- ✓ Does 0.16 mm layer height really save 30% time over 0.12 mm?
- ✓ Is tree support lighter than grid for this model?
- ✓ Does PETG burn meaningfully more filament than PLA at the same wall count?

Workflow

- 1 Open the Slicer dialog as usual. Pick your **baseline** configuration (Profile A) on the left.
- 2 Toggle **Compare engines** at the top of the dialog. A second panel (Profile B) appears next to A, pre-populated with A's settings so you only edit what you want to vary.
- 3 Change anything in B — layer height, walls, infill, supports, material, even the entire engine (Orca vs Cura vs Prusa).
- 4 Click **Slice Both**. Progress bars run in parallel; the slower engine determines total wait time.
- 5 Read the side-by-side **metrics panel** below. Differences are highlighted: green = better, red = worse. See section 4 for what each metric means.

4. Metrics — what every row means

Metric	Units	What it tells you
Print time	h:mm	Estimated wall-clock time. Most important number for batch production runs.
Filament length	m	How much filament is extruded. Useful for spool budgeting.
Filament weight	g	Mass of plastic used. Direct cost driver — multiply by your €/kg material price.
Peak temperature	°C	Hottest extrusion temp. High peaks for PA/CF blends; matters for hot-end longevity.
Layer count	—	Total layers in the slice. Time = layer count × mean layer time, roughly.
Support volume	cm ³	Plastic spent on supports. Tree supports usually score 30–50% lower here than grid.
Wall path length	m	Total perimeter extrusion. Watch for >3x ratios — usually means you cranked up walls/concentric infill.
Top/bottom area	cm ²	Surface area of solid top/bottom layers. Big differences here warn that part orientation changed.

Diff coloring: a metric is highlighted green when it's ≥5% better in B than A (lower-is-better for time/weight/supports). Red means ≥5% worse. Grey means within 5% — noise, not signal.

5. Compare-engine recipes

Speed-vs-quality A/B

Profile A: 0.12 mm layer, 4 walls, 25% gyroid infill.

Profile B: 0.20 mm layer, 3 walls, 15% gyroid infill.

Expect B to slice 35–50% faster and 10–15% lighter, with visibly more striation. Worth it for prototypes; not for showpieces.

Support style A/B

Profile A: tree (organic) supports, 30° threshold.

Profile B: grid supports, 30° threshold.

Tree usually wins on support weight by 30–50%; grid is more reliable on flat overhangs. Compare both, pick per geometry.

Material A/B

Profile A: Generic PLA at 215°C / 60°C.

Profile B: Generic PETG at 240°C / 80°C.

Same model, two filaments. Tells you the print-time delta (PETG ~5% slower due to lower max speeds) and the weight delta (PETG ~25% denser than PLA).

Wall count sweep

Profile A: 2 walls.

Profile B: 4 walls.

Wall count is the strongest single lever for part strength on functional prints. The metric panel will show you exactly how much extra time and filament that strength costs.

6. Send to your desktop slicer

Sometimes you want the full bells-and-whistles of a desktop slicer (custom plates, calibration shapes, advanced supports). ForgeSlicer can hand off in three formats:

Format	Best for
.3mf	OrcaSlicer / Bambu Studio. Preserves modifiers, color assignments, and per-object settings.
.stl	PrusaSlicer / Cura / generic. Loses modifier metadata but universally readable.
.gcode (preview)	Quick eyeball only — ForgeSlicer's internal preview engine, not production-quality.

7. Troubleshooting

Symptom	Likely cause	Fix
Slice fails with 'mesh not manifold'	Booleans left thin slivers or coincident faces.	Re-run booleans with 0.05 mm overlap, OR enable the slicer's auto-repair option.
Print time estimate is wildly off	Accel/jerk profile doesn't match your printer's firmware.	Update the system printer profile to match your machine's actual max speeds.

Compare shows huge weight delta but same volume	Materials have different densities (PLA 1.24, PETG 1.27, TPU 1.21, PA-CF 1.08).	Expected — the metric is mass, not volume.
Server slice keeps timing out	Very large mesh (>500 k tris) or very fine layer height.	Decimate the mesh first OR raise layer height to 0.20 mm for the first pass.

8. Quick reference

Toolbar: Send to Slicer button (orange) is always second from the right of Row 1.

Hotkey: Ctrl+P opens the Slicer dialog directly.

Compare toggle: top of the dialog, next to the engine picker.

Source files: `backend/orca_engine.py` routes CLI calls; `frontend/src/components/SlicerDialog.jsx` renders the side-by-side panel.

Profile docs: System profiles inherit from upstream OrcaSlicer — github.com/SoftFever/OrcaSlicer has the canonical parameter reference.

Tutorial generated May 29, 2026 from ForgeSlicer source. Run the corresponding `scripts/build*_tutorial.py` to regenerate after future updates.